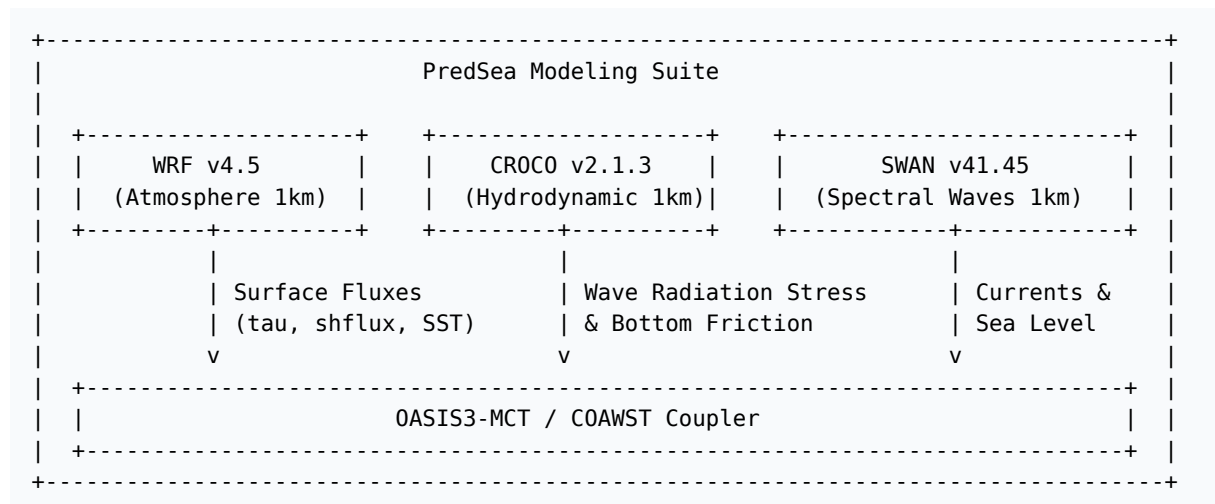


## 02. Core Numerical Engines & Coupling Mechanisms

This document provides a mathematical and functional analysis of the core numerical modeling engines integrated into the **PredSea** forecasting suite: **WRF** (atmospheric dynamics), **CROCO** (hydrodynamics), and **SWAN** (spectral wave dynamics), alongside their two-way and three-way coupling interfaces via the **OASIS3-MCT / COAWST** framework.

### 1. Core Numerical Modeling Engines



#### A. WRF (Weather Research and Forecasting Model)

The atmospheric component runs WRF v4.5, solving the fully compressible, non-hydrostatic Euler primitive equations on a Arakawa-C grid using terrain-following hydrostatic pressure vertical coordinates ( $\eta$ ).

- **Primary Governing Variables:** 3D velocity vectors ( $\vec{u}_a$ ), perturbation potential temperature ( $\theta'$ ), geopotential ( $\phi'$ ), and surface pressure ( $P_{atm}$ ).
- **Physical Parameterizations:**
  - **Microphysics:** WSM6 (WRF Single-Moment 6-class scheme).
  - **Planetary Boundary Layer (PBL):** YSU (Yonsie University scheme) resolving atmospheric turbulence and surface momentum flux closure.
  - **Radiation:** RRTMG longwave and shortwave schemes computing surface downward fluxes ( $SW_{\downarrow}, LW_{\downarrow}$ ).
- **Output Parameters:** Provides 10 m wind vectors ( $U_{10}, V_{10}$ ), 2 m air temperature ( $T_a$ ), specific humidity ( $q_a$ ), surface pressure ( $P_{atm}$ ), and radiative fluxes ( $SW_{\downarrow}, LW_{\downarrow}$ ).

#### B. CROCO (Coastal and Regional Ocean Community Model)

CROCO v2.1.3 is a free-surface, hydrostatic/non-hydrostatic 3D primitive equation hydrodynamic model evolved from ROMS. It uses an Arakawa-C grid in the horizontal and a general curvilinear, terrain-following  $s$ -vertical coordinate system in the vertical.

#### Hydrodynamic Governing Equations

In Cartesian/curvilinear coordinates with terrain-following  $s$ -levels, the Reynolds-averaged Navier-Stokes (RANS) momentum equations under the Boussinesq and hydrostatic approximations are:

$$\frac{\partial u}{\partial t} + \vec{v} \cdot \nabla u - f v = -\frac{1}{\rho_0} \frac{\partial p}{\partial x} + \frac{\partial}{\partial z} \left( K_m \frac{\partial u}{\partial z} \right) + \mathcal{D}_u$$

$$\frac{\partial v}{\partial t} + \vec{v} \cdot \nabla v + f u = -\frac{1}{\rho_0} \frac{\partial p}{\partial y} + \frac{\partial}{\partial z} \left( K_m \frac{\partial v}{\partial z} \right) + \mathcal{D}_v$$

$$\frac{\partial p}{\partial z} = -\rho g$$

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$$

Where: \*  $u, v, w$  are the 3D fluid velocity components in  $x, y, z$ . \*  $f = 2\Omega \sin \phi$  is the Coriolis parameter. \*  $\rho_0$  is the reference ocean water density (1025 kg/m<sup>3</sup>). \*  $K_m$  is the vertical eddy viscosity derived from GLS (Generic Length Scale)  $k$ - $\epsilon$  or  $k$ - $\omega$  turbulence closure. \*  $\mathcal{D}_u, \mathcal{D}_v$  represent horizontal viscosity and dissipation operator terms.

### Stretched $s$ -Vertical Coordinate System

To resolve both deep ocean circulation and shallow coastal boundary layers, CROCO employs a non-linear vertical transformation (`NEW_S_COORD`):

$$z(x, y, s) = \zeta(x, y) + [\zeta(x, y) + h(x, y)] \cdot S(x, y, s)$$

Where the non-linear stretching function  $S(x, y, s)$  is governed by parameters  $\theta_s$  (surface stretching),  $\theta_b$  (bottom stretching), and  $h_c$  (critical depth):

$$S(x, y, s) = \frac{h_c s + hC(s)}{h_c + h}$$

In the reference Balearic grid (401 × 501 horizontal grid at 1 km resolution),  $N = 32$  vertical layers are configured with  $\theta_s = 6.0$ ,  $\theta_b = 0.0$ , and  $h_c = 10$  m.

### C. SWAN (Simulating WAVes Nearshore)

SWAN v41.45 is a third-generation spectral wave model that computes the evolution of the 2D wave action density spectrum  $N(\sigma, \theta; x, y, t)$  over coastal and shelf sea environments:

$$N(\sigma, \theta) = \frac{E(\sigma, \theta)}{\sigma}$$

Where  $\sigma$  is the relative wave intrinsic frequency and  $\theta$  is the wave propagation direction.

### Spectral Action Balance Equation

The governing wave transport equation in absolute Cartesian coordinates is given by:

$$\frac{\partial N}{\partial t} + \frac{\partial}{\partial x}(c_x N) + \frac{\partial}{\partial y}(c_y N) + \frac{\partial}{\partial \sigma}(c_\sigma N) + \frac{\partial}{\partial \theta}(c_\theta N) = \frac{S_{tot}}{\sigma}$$

Where: \*  $(c_x, c_y) = \vec{c}_g + \vec{U}$  are the spatial propagation velocity components (group velocity  $\vec{c}_g$  plus background current vector  $\vec{U}$ ). \*  $c_\sigma, c_\theta$  represent the propagation speeds in spectral frequency  $\sigma$  and direction  $\theta$  (resolving current refraction and depth-induced shoaling). \*  $S_{tot}$  is the total source/sink term:

$$S_{tot} = S_{in} + S_{nl3} + S_{nl4} + S_{ds} + S_{bot} + S_{db}$$

Where  $S_{in}$  is wind input,  $S_{nl3}, S_{nl4}$  are 3-wave (triad) and 4-wave (quadruplet) non-linear interactions,  $S_{ds}$  is whitecapping dissipation,  $S_{bot}$  is bottom friction, and  $S_{db}$  is depth-induced wave breaking.

## 2. Inter-Model Exchange Dynamics (WRF - SWAN - CROCO)

The complete three-way feedback mechanism across WRF, SWAN, and CROCO (ROMS) is illustrated in the architectural figure below:

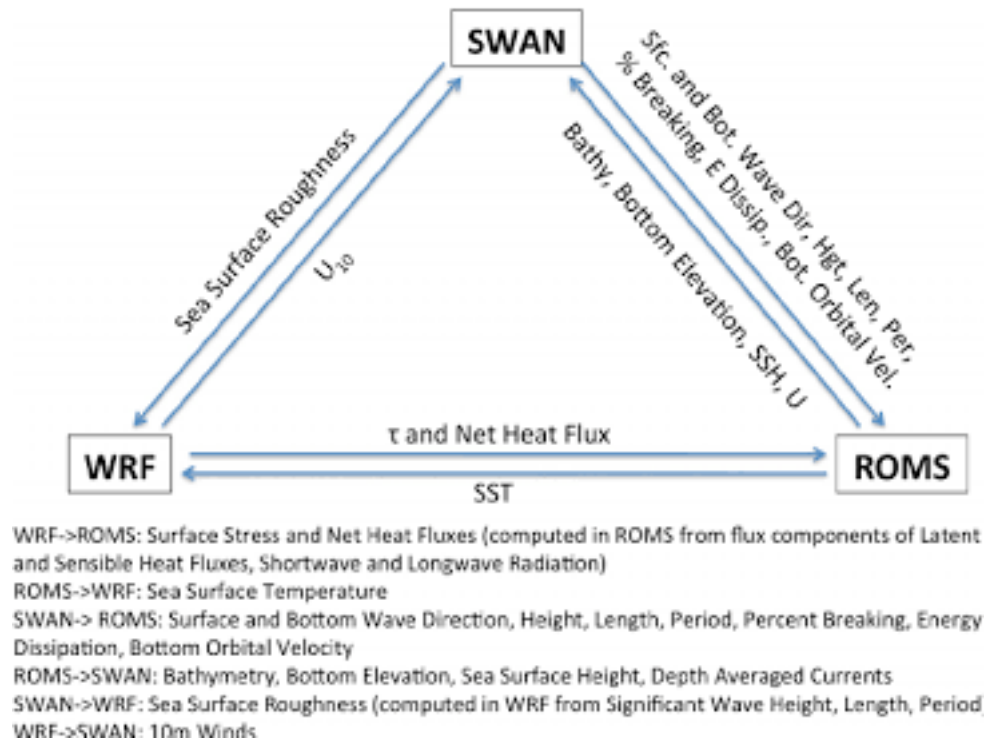


Figure 1: WRF-SWAN-CROCO Inter-Model Exchange Dynamics

Figure 2.1: Inter-model exchange dynamics within the COAWST framework, defining variable feedback paths between WRF (atmosphere), SWAN (waves), and CROCO/ROMS (hydrodynamics).

```

flowchart TD
    subgraph Atmosphere
        WRF["WRF (Weather Research & Forecasting)"]
    end

    subgraph Wave Dynamics
        SWAN["SWAN (Spectral Waves)"]
    end

    subgraph Hydrodynamics
        CROCO["CROCO / ROMS (Hydrodynamics)"]
    end

    %% WRF <-> CROCO
    WRF -- "Tau (Surface Stress) & Net Heat Flux" --> CROCO
    CROCO -- "Sea Surface Temp (SST)" --> WRF

    %% WRF <-> SWAN
    WRF -- "10m Wind Vectors (U10, V10)" --> SWAN
    SWAN -- "Sea Surface Roughness (z0)" --> WRF

    %% SWAN <-> CROCO
    SWAN -- "Wave Dir, Hgt, Len, Per, % Breaking, E_diss, Bot Orbital Vel" --> CROCO
    CROCO -- "Bathymetry, Bottom Elevation, Sea Level (Zeta), Currents (u,v)" --> SWAN

```

## Detailed Directional Exchange Vector Breakdown

### 1. WRF → CROCO (ROMS)

- **Surface Stress ( $\tau$ ) & Net Heat Flux:** WRF provides atmospheric surface stress vectors ( $\tau_x, \tau_y$ ) and component radiative/turbulent heat fluxes (`radsw`, `shflx_rlw`, `shflx_lat`, `shflx_sen`) to drive ocean surface momentum and mixed-layer thermodynamics in CROCO.

### 2. CROCO (ROMS) → WRF

- **Sea Surface Temperature (SST):** CROCO returns updated 1 km spatial SST fields back to WRF. This dynamic SST feedback prevents atmospheric boundary layer temperature drift and corrects surface sensible/latent heat transfer coefficients.

### 3. SWAN → CROCO (ROMS)

- **Wave Parameters & Bottom Kinematics:** SWAN transmits surface and bottom wave direction, significant wave height ( $H_s$ ), wavelength ( $L$ ), peak period ( $T_p$ ), percent wave breaking fraction, energy dissipation rate ( $E_{diss}$ ), and bottom orbital velocity ( $U_{bot}$ ) into CROCO. These drive wave radiation stress gradients ( $S_{xx}, S_{xy}, S_{yy}$ ) and enhance bottom boundary layer friction.

### 4. CROCO (ROMS) → SWAN

- **Hydrodynamic Conditions:** CROCO feeds updated bathymetry, bottom elevation changes, sea surface height ( $\zeta$ ), and 3D depth-averaged currents ( $u, v$ ) into SWAN. These adjust shallow-water shoaling limits, depth-induced wave breaking, and Doppler current refraction.

### 5. SWAN → WRF

- **Sea Surface Roughness ( $z_0$ ):** SWAN computes wave-age and steepness dependent aerodynamic surface roughness length ( $z_0$ ) from significant wave height, length, and period, passing it to WRF to adjust atmospheric drag coefficients ( $C_D$ ).

## 6. WRF → SWAN

- **Surface Wind Forcing ( $U_{10}, V_{10}$ ):** WRF passes high-resolution 10 m wind velocity vectors into SWAN to drive spectral wave growth ( $S_{in}$ ).

## 3. Summary of Coupler Data Exchange Matrix

| Source Model | Target Model | Exchange Variable               | Symbol / Units                                                  | Physical Coupling Effect                                           |
|--------------|--------------|---------------------------------|-----------------------------------------------------------------|--------------------------------------------------------------------|
| <b>WRF</b>   | <b>CROCO</b> | Surface Stress & Heat Flux      | $\tau, \text{shflux}$<br>(N/m <sup>2</sup> , W/m <sup>2</sup> ) | Drives Ekman currents & water column thermal structure             |
| <b>CROCO</b> | <b>WRF</b>   | Sea Surface Temperature         | SST (° C)                                                       | Modulates atmospheric boundary layer stability & flux coefficients |
| <b>SWAN</b>  | <b>CROCO</b> | Wave Height, Period & $U_{bot}$ | $H_s, T_p, U_{bot}$<br>(m, s, m/s)                              | Drives wave radiation stresses & bottom friction enhancement       |
| <b>CROCO</b> | <b>SWAN</b>  | Currents & Sea Surface Height   | $u, v, \zeta$ (m/s, m)                                          | Causes Doppler shift, wave refraction, & depth-induced breaking    |
| <b>SWAN</b>  | <b>WRF</b>   | Surface Roughness Length        | $z_0$ (m)                                                       | Adjusts atmospheric surface drag based on real wave state          |
| <b>WRF</b>   | <b>SWAN</b>  | 10 m Surface Wind Vectors       | $U_{10}, V_{10}$ (m/s)                                          | Governs spectral wave energy generation ( $S_{in}$ )               |